

Developmental variation in basal ganglia tissue iron, neurocognitive functioning, and impulsivity is associated with substance use trajectories in youth

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Neurodevelopmental models implicate dopaminergic and neurocognitive maturation in adolescent risk-taking, yet their joint contribution to substance use trajectories in humans remains unclear. We examined basal ganglia tissue iron, a marker of dopamine-related neurobiology, alongside impulsivity and inhibitory control in relation to longitudinal substance use patterns in the NCANDA-A cohort ($N = 802$; ages 12–30; 6,078 visits). Growth Mixture Models identified four trajectories: no/low use (30% of participants), youth peak (26%), adolescent increasing (17%), and adult increasing (26%). Substance use, inhibitory control, and tissue iron increased with age, while impulsivity declined. Greater substance use was associated with heightened impulsivity, low inhibitory control, and low tissue iron, prominently in early adolescence among youth peak patterns. Trajectories were further distinguished by divergent maturation of impulsivity and tissue iron. Results suggest that developmental variation in tissue iron and neurocognition contribute to youth substance use, highlighting adolescence as a sensitive window for risk stratification and prevention.

Illicit substance use increases dramatically during the teenage years. In 2024, only 15% of 8th graders in the United States reported using substances, compared to ~37% of 12th graders¹. Substance use is a leading contributor to preventable morbidity and mortality in youth, driving a range of behavioral and adverse health outcomes including suicidality and self-harm^{2,3}, interpersonal violence², and accidental death (i.e., road injuries² and drug-related toxicity^{4,5}). Accordingly, youth

substance use poses a critical public health concern, necessitating evidence-based policies and developmentally informed prevention and intervention strategies. Neurodevelopmental theories conceptualize youth substance use within dual systems frameworks, including the Driven Dual Systems Model^{6,7}. This model proposes that a developmental peak in subcortical, dopaminergically-mediated (DA) reward systems interacts with still-maturing, yet largely accessible,

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prefrontal cognitive control systems. As such, heightened reward reactivity can bias decision-making processes and contribute to normative increases in sensation seeking and risk-taking, creating a window of vulnerability for elevated substance use.

Specifically, ‘dual-risk’ theories of adolescent substance use propose that hyper-active striatal reward reactivity coupled with hypo-function of neurocognitive systems confers risk for problematic use^{8–11}, potentially via an increase of an already-enhanced reward drive¹² coupled with ongoing maturation of prefrontal cognitive control systems⁸. Empirical work supports this framework, with striatal reward hyper-activation identified as a reliable functional brain marker of adolescent substance use vulnerability^{8,13}, as well as studies showing that cognitive performance and cortical activation are associated with substance use risk-factors (i.e., impulsivity and externalizing^{14,15}) and substance use onset¹⁶. Normative adolescent increases in risk-taking have been linked to enhanced mesocorticolimbic functional connectivity, including ventromedial PFC (vmPFC) - ventral striatum (VS) coupling¹⁷, and VS connectivity predicts youth substance use onset and frequency^{18,19}. In adults, substance use is associated with impaired inhibitory control and distributed system-level alterations across cortical control regions (i.e., vmPFC, cingulate, insula, and dorsolateral PFC^{20,21}), potentially reflecting long-term dysregulation of top-down control over striatal DA^{22–25}. Together, these findings implicate mesocorticolimbic circuits underlying neurocognition and reward processing as a core feature of substance use vulnerability; however, how developmental interactions among DA neurobiology, cognitive control, and impulsivity shape adolescent substance use – and function as neurodevelopmental markers that predict its time course across adolescence into adulthood – remains unclear.

Animal studies demonstrate dynamic adolescent changes in DA systems, including increased striatal DA availability^{26,27} and signaling²⁸, supporting heightened reward drive, risk-taking^{27,29–31}, and drug-seeking behavior²⁸. Human fMRI studies similarly show amplified adolescent VS blood oxygen level dependent (BOLD) activation during reward processing^{12,32–34} that relate to risk-taking^{32,35}, and predict increases in substance use¹³. However, striatal BOLD is an indirect proxy for DA system function, and the lack of non-invasive *in vivo* markers in pediatric populations precludes understanding DA’s role in risk-taking and substance use behavior in youth. Molecular imaging studies in adults reveal distinct DAergic processes involved in substance use progression^{36–42}, but the few molecular imaging studies in youth have generally included small samples, primarily targeting young adults, and have yielded mixed results – with studies showing either no changes⁴³ or blunted DA function in high-risk youth⁴⁴. Based on these results, changes in DA neurobiology could reflect a long-term consequence of increased midbrain DA biosynthesis in response to substance use (i.e., downregulation of striatal DA signaling)²⁰, but findings in adults cannot disentangle preexisting vulnerability markers from long-term consequences of exposure. Animal work suggests that lower baseline levels of striatal D2 receptor availability and DA release promotes increased drug-seeking behavior in animals with no prior history of exposure^{25,45}, raising the possibility that these features reflect predisposing markers. To this end, longitudinal studies in substance naïve youth are essential to identify individual differences in DAergic processes prior to the onset of substance use, and to parse out neurodevelopmental markers that may differentiate long-term trajectories of substance use (i.e., escalation into adulthood) from adolescent peaks in substance use that reflect an instantiation of a normative risk-taking process.

Magnetic Resonance (MR)-based tissue iron neuroimaging provides a scalable, non-invasive index of basal ganglia DA-related physiology that can be obtained in pediatric cohorts. Normalized, time-averaged T2*-weighted (nT2*w) signal acquired during fMRI reflects the relative T2* relaxation across the brain and is sensitive to magnetic field inhomogeneities induced by iron^{46–48}. Tissue iron is involved in

several neurodevelopmental processes, and plays a central role in DA neurobiology and function: it localizes to dendrites of DA neurons⁴⁹, co-localizes with DA vesicles⁵⁰, and is involved in monoamine synthesis and production^{51–53} as a co-factor for tyrosine hydroxylase^{50,54}, the rate limiting enzyme in DA synthesis. Animal studies provide mechanistic and histological validation, showing strong correspondence between MRI-based iron measures (e.g., T2* relaxation) and histological iron content^{55–57}, and showing that iron availability regulates DA neurotransmission⁵⁸, receptor density^{59,60}, transporter function^{61,62}, and presynaptic vesicular integrity⁵⁰. The basal ganglia – a set of DAergically innervated subcortical nuclei including the striatum (caudate, putamen, and nucleus accumbens) and pallidum – and midbrain are among the most iron-rich brain regions^{63–66}, and MR indices correlate with post-mortem iron concentrations^{47,67} and PET [¹¹C]dihydrotetrabenazine (DTBZ)^{68,69} indices of presynaptic vesicular DA storage. Developmentally, tissue iron accrual increases throughout the first two decades of life and decelerates into adulthood^{70–74}, paralleling changes in DA availability observed in animal models³⁰. Emerging evidence links basal ganglia tissue iron to reward sensitivity⁷⁵, inhibitory control^{72,76,77}, and functional changes in mesocorticolimbic networks during adolescence¹⁷, and initial studies have demonstrated a role for tissue iron and related midbrain measures of DA neurophysiology (neuromelanin) in adolescent⁷⁸ and young adult substance use⁷⁹. However, these studies have not examined whether developmental variation in tissue iron differentiates longitudinal substance use trajectories.

We leveraged the National Consortium on Alcohol and Neurodevelopment in Adolescence and Adulthood (NCANDA-A), a large, longitudinal, high-risk sample of adolescents and young adults, the majority of whom were substance-naïve at baseline (i.e., limited or no history of substance use at the first timepoint)⁸⁰. Given substantial interindividual variability in the developmental timing and pace of change in substance use, cognitive, and neurobiological systems, cross-sectional or mean-level analyses may obscure meaningful developmental heterogeneity. Therefore, we additionally employed a person-oriented longitudinal approach using growth mixture modeling to identify distinct within-person trajectories of substance use, rather than assuming a single normative pattern across the population. This approach allowed us to examine how both between-person differences and within-person trajectories of substance use correspond to variation in three developmentally sensitive markers that have been implicated in adult substance use – impulsivity, inhibitory control, and tissue iron indices of DA neurobiology – providing a more developmentally precise account of risk-related processes. Based on evidence that low baseline DA promotes drug seeking behavior^{25,45}, and that developmental increases in basal ganglia tissue iron support maturation of mesocorticolimbic circuitry and age-related reductions in risk-taking¹⁷, we hypothesized that lower basal ganglia tissue iron, in conjunction with elevated impulsivity and reduced inhibitory control, would predict greater substance use across development. Primary analyses focused on aggregated substance use (across nicotine, cannabis, and alcohol (including binge drinking)) to capture general risk-taking propensity, and secondary analyses examined substance-specific effects. By integrating longitudinal sampling of neurobiological and behavioral processes, we demonstrate that developmental variation in impulsivity, inhibitory control, and DAergic processes plays a key role in the timing and progression of substance use during the transition from adolescence to adulthood.

Results

We examined how non-invasive neuroimaging markers of dopamine (DA) – related neurobiology and neurocognitive functioning relate to substance use trajectories from adolescence into adulthood in the National Consortium on Alcohol and Neurodevelopment in Adolescence and Adulthood (NCANDA-A) cohort, which combines yearly

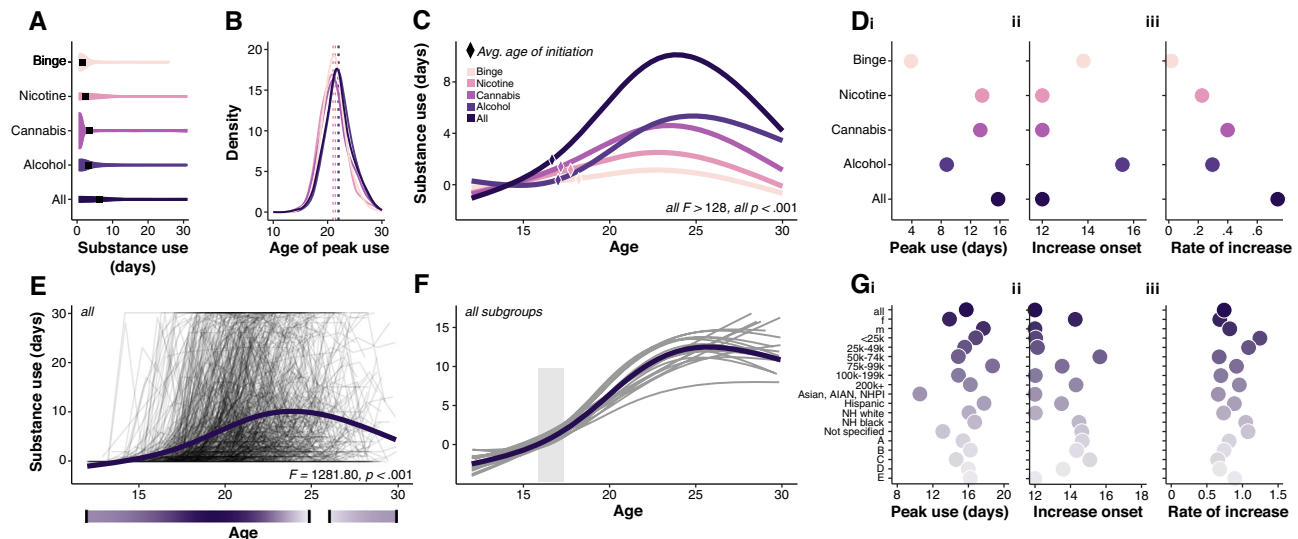


Fig. 1 | Patterns and developmental changes in substance use across the full sample and population subgroups. **A** Violin plots show distribution of self-reported past 30-day substance use across individual substances and a composite measure across all substances ('All') for all participants ($N = 802$; 6078 total visits). Black squares depict mean use. Color in (A) defines the key for substances in (B–D). **B** Histograms show age of peak use across individual substances. Mean peak age is depicted by dashed lines. **C** Developmental trajectories for substance use across all visits. Each line represents an individual substance, and the dark line represents composite use across all substances. Fit lines are from GAMM models (via *mgcv*⁶⁴, all $s(\text{age})$ $p < .001$ assessed via approximate F-tests). Average age of initiation across individual substances is represented by diamonds. **D** Circles represent **i.** Average peak use across substances (mean of per-participant peak value reflecting the maximum across visits), **ii.** Average age of increase onset across substances (first age where confidence interval of the derivative did not include zero, $p < .05$, two-sided), **iii.** Rate of increase across substances (calculated as the average first

derivative of the age spline). **E** Developmental trajectory for composite substance use (the principal metric for analyses), with black lines connecting visits for each participant. Fit lines are from GAMM models (via *mgcv*⁶⁴, $s(\text{age})$ $p < .001$ assessed via approximate F-tests). Corresponding color bars show rate of change during significant developmental periods (age-range where confidence interval of the derivative did not include zero). **F, G** Corresponding trajectories (F), average peak substance use (Gii), average age of increase onset (Gii), and rate of increase (Giii) for composite substance use across population subgroups, including 18 subgroups comprising sex, household income, race/ethnicity, and study site, analyzed individually. In (F), the dark line represents the full sample, and grey lines reflect individual subgroups. The shaded section reflects the range of age of initiation in composite substance use across subgroups. In (G – Giii), color indicates individual population subgroups. Average (Avg.). Female (F). Male (M). American Indian and Alaska Native (AIAN). Native Hawaiian and Pacific Islander (NHPI). Non-Hispanic (NH). Source data are provided as source data file 'Fig. 1'.

longitudinal neuroimaging with multidimensional assessments of executive function, trait impulsivity, and substance use. The sample included 807 youth (50% female) ages 12–30 across 1–9 longitudinal timepoints, for a total of 6164 sessions. Most participants were substance-naïve at baseline (with 83% reporting limited or no history of substance use at the first timepoint), and approximately 50% were recruited based on subclinical risk for alcohol use disorder⁸⁰. Self-report substance use was assessed at each visit, including quantity (number of days of use in the past 30 days), frequency ('never', 'ever', or 'regular (weekly)'), and polysubstance use (no-use, single-use, co-use (2 substances), or poly-use (> 2 substances)). Primary analyses examined generalized use patterns across all substances via a composite score; however, we also present substance-specific effects in the Supplement. We assessed impulsivity using the UPPS-P Impulsive Behavior Scale^{81–83}, inhibitory control using the well-validated rewarded anti-saccade task^{15,33,84}, and basal ganglia DA-related neurobiology⁶⁹ using MR-based estimates of tissue iron, obtained by time-averaged and normalized T2*-weighted imaging (nT2*w)^{73,74}. Generalized Additive Mixed Models (GAMMs) were used to estimate flexible, data-driven developmental trajectories for each measure, and to examine relationships between substance use and impulsivity, inhibitory control, and DA-related neurobiology, controlling for non-linear age effects (see Methods). A series of participant-level socio-demographic sensitivity analyses explored how the magnitude, age-related trajectories, and developmental timing of substance use, neurocognitive, and DAergic metrics, varied as a function of 18 subgroups defined by sex, race/ethnicity, study site, and socioeconomic indicators. Finally, growth mixture models were used to identify distinct

within-person trajectories in substance use across time, and GAMMs tested for individual differences neurocognitive and DAergic metrics across trajectory groups, and how developmental variation in these factors predicted patterns of youth substance use behavior. Full model statistics are provided in Supplementary Results and Tables.

Substance use increases from adolescence to young adulthood across the full sample

Across the full sample, use varied by substance ($F = 293.60$, $p < .001$), with highest use (Fig. 1A) and peak magnitudes (Fig. 1Di) for cannabis, followed by alcohol, nicotine, and finally, binge drinking. General substance use increased non-linearly until the mid-twenties, followed by declines into the 30s (Fig. 1C & E; composite use: $s(\text{age})$: edf = 2.98, $F = 785.07$, $p < .001$, $\text{partial } R^2 = .16$, model predicted change of -15 days, 50% of observed range in use). Though age-related changes were evident across each substance (Fig. 1C; $p < .001$ in all 4 models), substance-specific trajectories varied by shape (Fig. 1C), age of initiation (Fig. 1C), increase onset (Fig. 1Dii), and rates of change (Fig. 1Diii). Cannabis and alcohol use showed earlier initiation and steeper age-related increases, while binge drinking and nicotine showed later initiation and shallower, more gradual age-related increases. Finally, while alcohol use (including binge drinking) showed declines toward the end of the study period, cannabis and nicotine use stabilized at approximately 24 years of age. Peak use occurred around age 21–22 across substances (Fig. 1B). See Tables S1A and S2A for full model statistics across substances.

Across sociodemographic subgroups, overall magnitude (including peak levels) differed (Fig. 1Gi), but developmental timing was

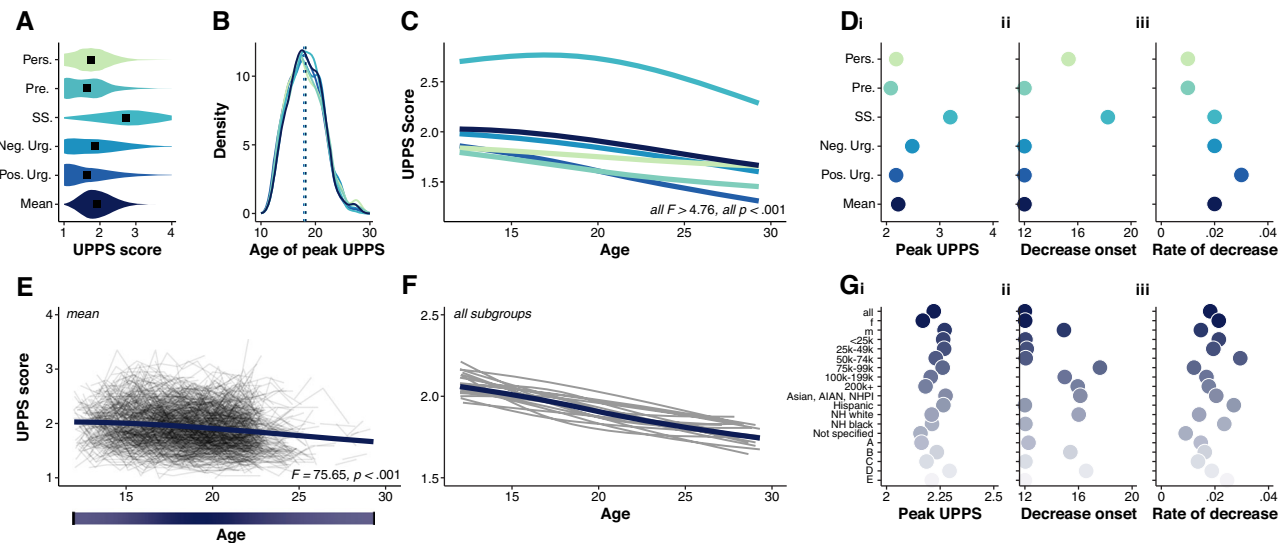


Fig. 2 | Patterns and developmental changes in impulsivity across the full sample and population subgroups. **A** Violin plots show distribution of self-reported impulsivity, measured by the Urgency-Premeditation-Perseverance-Sensation Seeking-Positive Urgency Impulsive Behavior Scale (UPPS), across individual subscales and the mean across all subscales for all participants ($N=802$; 4850 total visits). Black squares depict mean scores. Color in (A) defines the key for subscales in (B–D). **B** Histograms show age of peak UPPS across individual subscales. Mean peak age is depicted by hashed lines. **C** Developmental trajectories for UPPS scores across all visits. Each line represents an individual subscale, and the dark line represents the mean across all subscales. Fit lines are from GAMM models (via *mgcv*¹⁶⁴, all $s(\text{age})$ $p < .001$ assessed via approximate F-tests). **D** Circles represent **i.** Average peak score across subscales (mean of per-participant peak value reflecting the maximum across visits). **ii.** Average age of decrease onset across subscales (first age where confidence interval of the derivative did not include zero, $p < .05$, two-sided). **iii.** Rate of decrease across subscales (calculated as the average

first derivative of the age spline). **E** Developmental trajectory for mean UPPS scores (the principal metric for analyses), with black lines connecting visits for each participant. Fit lines are from GAMM models (via *mgcv*¹⁶⁴, $s(\text{age})$ $p < .001$ assessed via approximate F-tests). Corresponding color bars show rate of change during significant developmental periods (age-range where confidence interval of the derivative did not include zero). **F, G** Corresponding trajectories (F), average peak UPPS score (Gi), average age of decrease onset (Gii), and rate of decrease (Giii) for mean UPPS scores across population subgroups, including 18 subgroups comprising sex, household income, race/ethnicity, and study site, analyzed individually. In (F), the dark line represents the full sample, and grey lines reflect individual subgroups. In (Gi – Giii), color indicates individual population subgroups. Perseverance (Pers.). Premeditation (Pre.). Sensation Seeking (SS.). Negative Urgency (Neg. Urg.). Positive Urgency (Pos. Urg.). Female (F). Male (M). American Indian and Alaska Native (AIAN). Native Hawaiian and Pacific Islander (NHPI). Non-Hispanic (NH). Source data are provided as source data file ‘Fig. 2’.

relatively consistent (Fig. 1F), including age of initiation (Fig. 1F; ~15.5–17.5 years of age), increase onsets (Figure 1Gii; ~12–15 years), peak ages (~20.5–22.5 years), and comparable rates of change (Figure 1Giii). See Supplemental Results and Tables S3A–S6A for full model statistics.

Impulsivity decreases from adolescence to young adulthood across the full sample

Across the full sample, impulsivity varied by subscale ($F=3434.83$, $p < .001$), with highest scores (Fig. 2A) and peak scores (Fig. 2Di) for sensation seeking, followed by negative urgency, perseverance, positive urgency, and finally, premeditation. Mean impulsivity decreased non-linearly into the 30 s (Fig. 2C, E; mean UPPS: $s(\text{age})$: $\text{edf} = 2.42$, $F=75.66$, $p < .001$, $\text{partial } R^2 = .03$, model predicted change of $-.32$ points, 12.5% of observed range in UPPS). Though age-related changes were present across all subscales (Fig. 2C; $p < .001$ in all 5 models), subscale-specific trajectories varied by shape (Fig. 2C), decrease onsets (Fig. 2Dii), and rates of change (Figure 2Diii). Positive urgency, premeditation, and negative urgency began decreasing around age 12, whereas perseverance and sensation seeking showed protracted decrease onsets at approximately age 15 and 18, respectively. Finally, while all subscales showed stabilization at approximately 22–27 years of age, sensation seeking showed ongoing decreases toward the end of the study period. Peak impulsivity occurred around age 18 across subscales (Fig. 2B; see Tables S1B and S2B).

Across sociodemographic subgroups, overall magnitude (including peak magnitude) differed (Fig. 2Gi), but developmental timing was relatively consistent (Fig. 2F), including peak ages (~17.5–19 years of age) and comparable rates of change (Fig. 2Giii), but variation in

decrease onset ages (Figure 2Gii; see Supplemental Results and Tables S3B–S6B).

Inhibitory control performance increases from adolescence to young adulthood across the full sample

Inhibitory control was assessed using the anti-saccade task, a visuospatial task in which participants are presented with a peripheral cue and required to direct their gaze to its mirror location, requiring the suppression of the reflexive response to look toward the stimulus and the generation of a voluntary eye movement in the opposite direction (Fig. 3A; see Methods). Note that the anti-saccade task was administered at two participating research sites with a reduced sample size. Accuracy was measured as the percentage of correct trials, with 50% of trials rewarded (‘rewarded’ trials) and 50% non-rewarded (‘neutral’ trials).

Across the full sample, anti-saccade performance varied by trial type ($F=46.85$, $p < .001$), with highest performance (Fig. 3B) and peak performance (Fig. 3Di) for reward trials in comparison to neutral trials. Mean performance demonstrated inverse-shaped age-related increases beginning at approximately 12 years of age, followed by stabilization at approximately 20 years of age (Fig. 3C, E; mean anti-saccade: $s(\text{age})$: $\text{edf} = 1.90$, $F=14.82$, $p < .001$, $\text{partial } R^2 = .06$, model predicted change of -17% , 29% of observed range in performance). Age-related changes were evident across all trial types (Fig. 3C; $p < 2e-16$ in both models), and trial-type specific trajectories did not vary by shape (Fig. 3C), increase onsets, age of maturation (Fig. 3Dii), age of peak performance, or rates of change (Fig. 3Diii). We show age of maturation instead of age of increase onset given common increase onsets

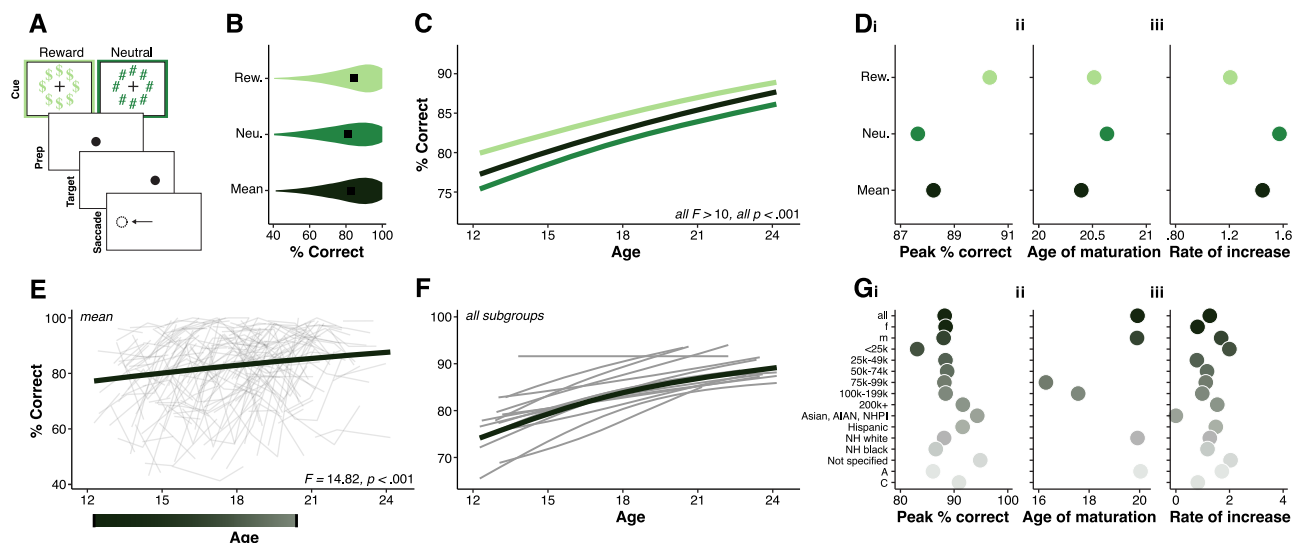


Fig. 3 | Patterns and developmental changes in inhibitory control across the full sample and population subgroups. **A** Anti-Saccade task. Trials began with a fixation cue surrounded by dollar ('reward') or pound ('neutral') signs, followed by a response preparation period. A peripheral stimulus then appeared along the horizontal midline, and participants were instructed to direct their gaze to its mirror location, completing an anti-saccade. **B** Violin plots show distribution of anti-saccade performance (% correct) across individual trial types and the mean across trial types for participants at the Duke and Pittsburgh sites ($N = 257$; 586 total visits). Black squares depict mean performance. Color in (B) defines the key for trial types in (C–D). **C** Developmental trajectories for performance across all visits. Each line represents an individual trial type, and the dark line represents the mean across all trials. Fit lines are from GAMM models (via *mgcv*¹⁶⁴, all $s(\text{age}) p < .001$ assessed via approximate F-tests). **D** Circles represent **i.** Average peak performance across trial types (mean of per-participant peak value reflecting the maximum across visits). **ii.** Average age of maturation across trial types (first age where confidence interval of the derivative included zero, $p < .05$, two-sided). **iii.** Rate of increase across trial

types (calculated as the average first derivative of the age spline). **E** Developmental trajectory for mean performance (the principal metric for analyses), with black lines connecting visits for each participant. Fit lines are from GAMM models (via *mgcv*¹⁶⁴, $s(\text{age}) p < .001$ assessed via approximate F-tests). Corresponding color bars show rate of change during significant developmental periods (age-range where confidence interval of the derivative did not include zero). **F**, **G** Corresponding trajectories (F), average peak performance (Gi), average age of maturation (Gii), and rate of increase (Giii) for mean performance across population subgroups, including 15 subgroups comprising sex, household income, race/ethnicity, and study site (note only two participating sites), analyzed individually. In (F), the dark line represents the full sample, and grey lines reflect individual subgroups. In (Gi–Giii), color indicates individual population subgroups. Reward (Rew). Female (F). Male (M). American Indian and Alaska Native (AIAN). Native Hawaiian and Pacific Islander (NHPI). Non-Hispanic (NH). Source data are provided as source data file 'Fig. 3'.

across trial types and given that the age at which inhibitory control reaches its developmental plateau is more insightful (see Tables S1C and S2C).

Across sociodemographic subgroups, overall magnitude (including peak magnitude) differed (Fig. 3Gi), but developmental timing was relatively consistent (Fig. 3F), including peak ages (~16.5–18.5 years of age) and comparable rates of change (Fig. 3Giii), but variation in age of maturation (Figure 3Gii; see Supplemental Results and Tables S3C–S6C).

Basal ganglia tissue iron decreases from adolescence to young adulthood across the full sample

MR-based indices of basal ganglia tissue iron, reflecting DA-related neurobiology⁶⁹ were obtained via time-averaged and normalized T2*-weighted imaging ($nT2^*w$) across an 10-minute resting state scan. Individual estimates were extracted across each basal ganglia region of interest, including the caudate nucleus, nucleus accumbens (NAcc), pallidum, and putamen, as well as one estimate reflecting the mean across all voxels in the basal ganglia (Fig. 4A; see Methods, note that $nT2^*w$ is inversely related to iron content).

Across the full sample, as expected, $nT2^*w$ varied by region ($F = 11281.35$, $p < .001$), with lowest $nT2^*w$ (Fig. 4B) and lowest peak $nT2^*w$ (Fig. 4Di), indicating highest iron, for the NAcc, followed by the pallidum, the caudate, and finally the putamen. Basal ganglia $nT2^*w$ decreased linearly (Fig. 4C, E; mean across all voxels: $s(\text{age})$: $\text{edf} = 2.70$, $F = 253.25$, $p < .001$, $\text{partial } R^2 = .07$, model predicted change of $\sim .001$ units, 27% of observed range in $nT2^*w$). Age-related changes were present across all regions (Fig. 4C; $p < .001$ in all 4 models), however, trajectories varied by shape (Fig. 4C), decrease onsets (Fig. 4Dii), rates

of change (Fig. 4Diii), and age of peak $nT2^*w$. Relative to other regions, the caudate showed shallower, later, and more gradual age-related decreases over time beginning around age 14, while the NAcc showed steeper, more rapid non-linear age-related decreases, prominently early in adolescence (with age of peak change occurring around 13), in comparison to other regions (with age of peak change occurring around 19–20). All regions showed peak $nT2^*w$ values in young adulthood around age 19–20, and continued to show age-related decreases, reflecting increases in iron accumulation, toward the end of the study period (see Tables S1D and S2D).

Across sociodemographic subgroups, overall magnitude (including peak magnitude) differed (Fig. 4Gi), but developmental timing was relatively consistent (Fig. 4F), including peak ages (~19.5–21 years of age) and comparable rates of change (Figure 4Giii), with some variation in decrease onset (indicating increase onset) ages (Figure 4Gii; see Supplemental Results and Tables S3D–S6D).

Together, these analyses show robust, generalizable patterns of longitudinal change in substance use, trait-level impulsivity, inhibitory control, and basal ganglia tissue iron across adolescence and young adulthood.

Increased substance use is associated with heightened impulsivity, decreased inhibitory control, and reduced basal ganglia tissue iron

To test whether neurocognitive and neurobiological measures tracked differences in substance use patterns, we modeled impulsivity, inhibitory control, and tissue iron across frequency (i.e., never, ever, regular) and use-type (i.e., no-use, single-use, co-use, poly-use) categories at each visit (see Table S7A for group characteristics across all

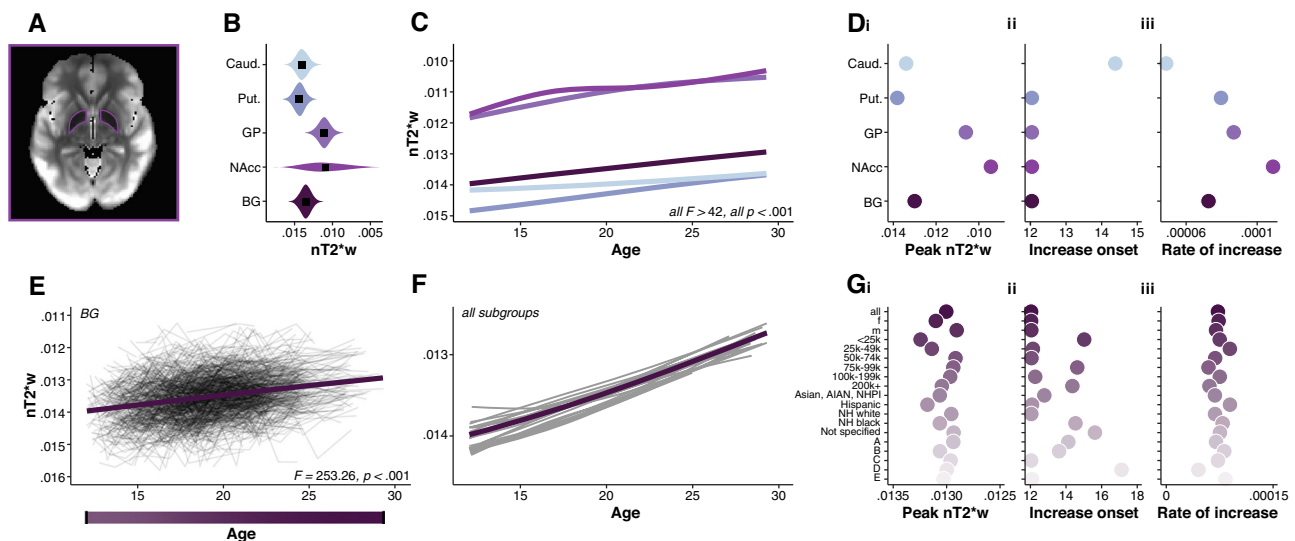


Fig. 4 | Patterns and developmental changes in tissue iron reflecting dopamine-neurobiology across the full sample and population subgroups. **A** Time-averaged and normalized T2*-weighted (nT2*w) signal across visits and participants. **B** Violin plots show distribution of nT2*w across individual regions, and the mean across all basal ganglia voxels for all participants ($N = 799$; 4328 total visits). Black squares depict mean nT2*w. Color in (B) defines the key for regions in (C,D). **C** Developmental trajectories for nT2*w across all visits. Each line represents an individual region, and the dark line represents the basal ganglia. Fit lines are from GAMM models (via *mgcv*¹⁶⁴, all $s(\text{age}) p < .001$ assessed via approximate F-tests). **D** Circles represent i. Average minimum nT2*w across ROIs (i.e., peak tissue iron; mean of per-participant value reflecting the minimum across visits). ii. Average age of nT2*w decrease onset (i.e., tissue iron increase onset) across ROIs (first age where confidence interval of the derivative did not include zero, $p < .05$, two-sided). iii. Rate of decrease in nT2*w (i.e., increase in tissue iron), across ROIs (calculated as the average first derivative of the age spline). **E** Developmental trajectory for basal ganglia nT2*w (the principal metric for analyses), with black lines connecting visits

for each participant. Fit lines are from GAMM models (via *mgcv*¹⁶⁴, $s(\text{age}) p < .001$ assessed via approximate F-tests). Corresponding color bars show rate of change during significant developmental periods (age-range where confidence interval of the derivative did not include zero). **F**, **G** Corresponding trajectories (F), average minimum nT2*w (Gi), average age of decrease onset (Gii), and rate of decrease (Giii) for basal ganglia nT2*w across population subgroups, including 18 subgroups comprising sex, household income, race/ethnicity, and study site, analyzed individually. In (F), the dark line represents the full sample, and grey lines reflect individual subgroups. In (Gi – Giii), color indicates individual population subgroups. Note that given inverse relationship between nT2*w and tissue iron x-axes in (B), (Di), (Dii), (Gi), (Gii), and y-axes in (C), (E), and (F) are reversed. Time-averaged and normalized T2*-weighted (nT2*w). Caudate (Caud.). Putamen (Put.). Globus Pallidum (GP). Nucleus Accumbens (NAcc). Basal ganglia (BG). Female (F). Male (M). American Indian and Alaska Native (AIAN). Native Hawaiian and Pacific Islander (NHPI). Non-Hispanic (NH). Source data are provided as source data file ‘Fig. 4’.

substances). Models further tested for associations with past 30-day substance use. All models controlled for a smoothed term of age, clarifying these relationships above and beyond age effects.

Impulsivity: Mean UPPS-P differed across substance use frequency categories (Fig. 5A top panel; $F = 24.63$, $p < .001$, $\text{partial } R^2 = .04$, model predicted difference of $-.11$ points, 4% of observed range in UPPS), and substance use-type categories (Fig. 5A bottom panel; $F = 18.33$, $p < .001$, $\text{partial } R^2 = .05$, model predicted difference of $-.10$ points, 4% of observed range in UPPS), with heightened scores scaling with frequency of use (regular > ever > never, all $p < .001$), and with poly-substance use engagement (poly-substance use > co-use > single-use > no-use, all $p < .001$). Increased impulsivity was further associated with greater past 30-day substance use (Fig. 5B; $s(\text{impulsivity})$: $\text{edf} = 1.62$, $F = 33.54$, $p < 2e-16$, $\text{partial } R^2 = .03$, model predicted change of $-.4$ days, 14.5% of observed range in use). Effects were consistent across UPPS-P subscales and substances (see Methods for lack of interaction effects and Tables S7Bi, Di).

Inhibitory Control: Mean anti-saccade performance differed across substance use frequency categories (Fig. 5C top panel; $F = 5.11$, $p = .006$, $\text{partial } R^2 = .003$, model predicted difference of $-.4\%$, 7% of observed range in performance), and substance use-type categories (Fig. 5C bottom panel; $F = 5.22$, $p = .001$, $\text{partial } R^2 = .003$, model predicted difference of $-.5\%$, 8.5% of observed range in performance), with decreased performance broadly scaling with frequency of use (regular > ever and never, all $p < .01$), and with poly-substance use engagement (poly-substance use > no-use, $p < .02$). In contrast, we did not observe a significant association between mean anti-saccade performance and past 30-day substance use (Fig. 5D; $s(\text{inhibitory control})$: $\text{edf} = 1.00$, $F = .48$, $p = .49$, $\text{partial } R^2 = .0008$, model predicted change of <1 day,

2.5% of observed range in use). Effects were similar across anti-saccade trial types (see Methods for lack of interaction effects and Tables S7Bii, Dii).

Tissue Iron: Basal ganglia nT2*w differed across substance use frequency categories (Fig. 5E top panel; $F = 4.22$, $p = .01$, $\text{partial } R^2 = .0005$, model predicted difference of $-.001$ units, 2% of observed range in nT2*w), with higher nT2*w, indicating lower iron, broadly scaling with frequency of use (regular > ever and never, all $p < .01$). Higher nT2*w (lower iron) was further associated with greater past 30-day substance use (Fig. 5F; $s(\text{nT2*w})$: $\text{edf} = 1.71$, $F = 3.37$, $p = .02$, $\text{partial } R^2 = .003$, model predicted change of $-.2$ days, 7% of observed range in use). nT2*w did not differ significantly across substance use-type categories (Fig. 5E bottom panel; $F = 1.14$, $p = .33$, $\text{partial } R^2 = .0006$, model predicted difference of $<.001$ units, 1% of observed range in nT2*w). Effects were similar across basal ganglia ROIs (see Methods for lack of interaction effects and Tables S7Biii–S7Diii).

Together, these results suggest that increased substance use was associated with heightened impulsivity, decreased inhibitory control, and lower basal ganglia tissue iron.

Characterizing within-person patterns of substance use

Growth Mixture Models (GMMs) were applied to capture variation in within-person trajectories (patterns of change) across timepoints and classify individuals based on generalized patterns of overall substance use (aggregate across all substances) across development.

GMMs revealed four classes of trajectories in substance use across the full sample (Fig. 6A, top panel), all of which showed significant age-related increases in substance use ($p < .001$ in all 4 models). However, trajectory groups differed in the magnitude of substance use (Fig. 6B;

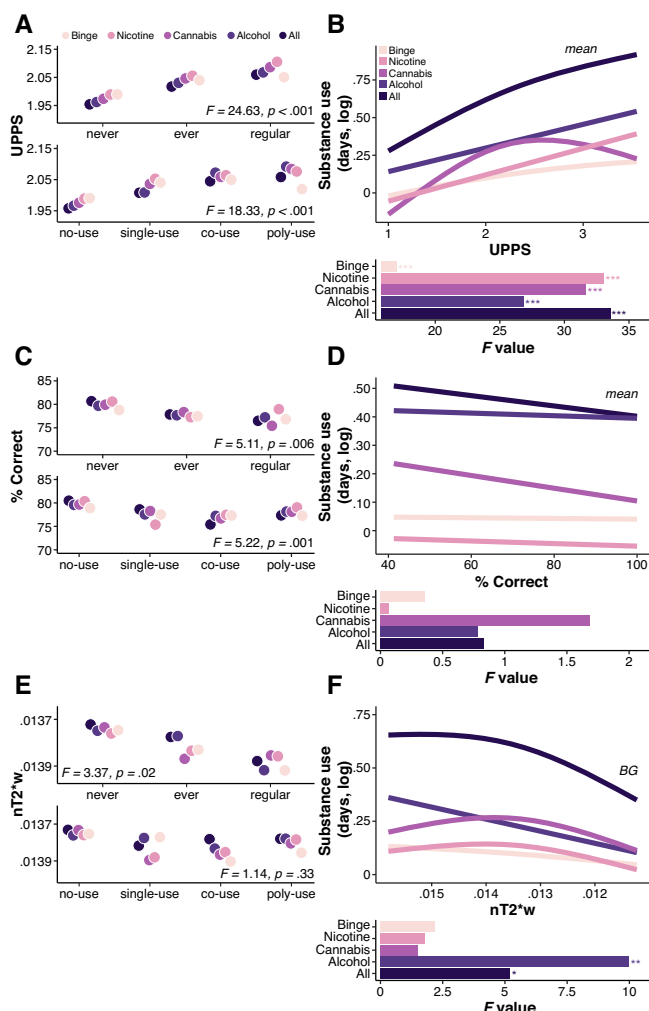


Fig. 5 | Associations between substance use and impulsivity, inhibitory control, and tissue iron across the full sample. **A** GAMM-predicted mean UPPS scores (via *ggpredict*⁶⁵) within substance use frequency (top panel) and use-type (bottom panel) categories. **B** GAMM fits from non-linear models assessing associations between substance use and mean UPPS scores for all participants ($N = 802$; 4850 total visits). **C** GAMM-predicted mean anti-saccade performance within substance use frequency (top panel) and use-type (bottom panel) categories. **D** GAMM fits from non-linear models assessing associations between substance use and mean anti-saccade performance for participants at the Duke and Pittsburgh sites ($N = 257$; 586 total visits). **E** GAMM-predicted basal ganglia $nT2^*w$ within substance use frequency (top panel) and use-type (bottom panel) categories. **F** GAMM fits from non-linear models assessing associations between substance use and $nT2^*w$ for all participants ($N = 799$; 4328 total visits). In (A), (C), and (E), circles in the top panel reflect substance-specific categorizations based on whether participants reported never having used each substance ('never'), ever having used each substance ('ever'), and regularly using each substance (> 1 /week) at each visit. Circles in the bottom panel reflect substance-specific categorizations based on whether participants reported no-use (i.e., never), use of each substance ('single-use'), the use of each substance in addition to one other substance ('co-use'), or in addition to more than two other substances ('poly-use') at each visit. Statistics in (A), (C), and (E) represent a main effect of the 'All' category (regardless of which substance was endorsed) from an ANOVA test (two-tailed), controlling for age and sociodemographic variables. In (B), (D), and (F), y-axes in the top panels show log-transformed past 30-day substance use, each fit line represents an individual substance, and the dark line represents composite use across all substances. The bottom panels in (B), (D), and (F) show bars reflecting F-values and stars denoting significance of the smooth term for each independent variable for each substance (* $p < .05$, ** $p < .01$, *** $p < .001$, uncorrected). Note that the y-axis in (E) and the x-axis in (F) are reversed such that visually higher values reflect higher tissue iron concentration (lower $nT2^*w$). Urgency-Premeditation-Perseverance-Sensation Seeking-Positive Urgency Impulsive Behavior Scale (UPPS). Time-averaged and normalized $T2^*w$ ($nT2^*w$). Source data are provided as source data file 'Fig. 5'.

$F = 113.88$, $p < .001$, $\text{partial } R^2 = .23$, model predicted difference of ~ 7 days, 24% of observed range in use), and in the shape (Fig. 6A), age of initiation (Fig. 6A), age of peak use (Fig. 6C), increase onsets (Figure 6Dii), and rate of change (Figure 6Diii). The low group ($n = 241$, 30% of participants) showed minimal increases across adolescence, with near-zero growth until approximately 22 years of age. The youth peak group ($n = 211$, 26% of participants) showed the earliest initiation and highest level of substance use, with a quadratic pattern marked by steep, rapid age-related increases beginning around age 12, followed by declines beginning around age 26, ultimately converging toward low-use levels by young adulthood. In contrast, the adolescent increasing group ($n = 139$, 17% of participants) showed steep, linear increases beginning around age 16. Finally, the adult increasing group ($n = 211$, 26% of participants) showed slower, later, and more gradual linear increases continuing until the end of the study period, and a later age of peak use (around age 23) in comparison to other groups, which peaked earlier (around age 21–22). These patterns held across each individual substance (see Tables S8A–S11A).

Sensitivity analyses confirmed that group membership was not significantly associated with variation in sociodemographic factors (Fig. 6A, bottom panel; i.e., no sociodemographic factors were disproportionately represented within a particular trajectory category). Despite a significant effect of age on trajectory groups ($F = 19.90$, $p < .001$), there was only minor variation (ranging from mean ages of 18.8–20.5 years).

We next investigated individual differences across trajectory groups, as well as developmental variation in these factors across trajectories that might indicate that individual differences occurred early in development (i.e., prior to substance use onset) or later in development (i.e., following substance exposure).

Distinct developmental patterns of impulsivity across trajectory groups

Mean UPPS-P differed across trajectory groups (Fig. 7B, Di; $F = 14.13$, $p < .001$, $\text{partial } R^2 = .04$, model predicted difference of $\sim .20$ points, 7% of observed range in UPPS), with highest impulsivity in the youth peak group, followed by the adolescent increasing group, the low group, and lastly, the adult increasing group. Age-related decreases were evident across all groups (Fig. 7A; $p < .001$ in all 4 models), however, trajectories differed in shape (Fig. 7A), decrease onsets (Fig. 7Cii), and rate of change (Figure 7Ciii). The low, youth peak, and adult increasing groups showed steep, early decreases beginning early in adolescence and persisting until the mid-to-late twenties. In contrast, the adolescent increasing group showed later, shallower, and slower declines (see Tables S8B–S11B).

We observed a significant trajectory group by age interaction on impulsivity ($F = 9.89$, $p < .001$), and post-hoc tests revealed that in early adolescence (estimates for age 14, see methods, Fig. 7D top panel), prior to the mean age of SU initiation, the youth peak group demonstrated significantly higher impulsivity in comparison to all other trajectory groups (all $p < .01$), and the adolescent increasing group showed significantly higher scores in comparison to the adult increasing group ($p = .03$). By young adulthood (estimates for age 25, see methods, Fig. 7D top panel), UPPS-P scores remained higher in the youth peak and adolescent increasing groups (all $p < .001$). Additionally, while the youth peak group showed comparable slopes to the low and adult increasing groups, potentially reflecting normative decreases in impulsivity over time, the adolescent increasing group showed significantly flatter slopes (Fig. 7D bottom panel), indicating diminished age-related decreases in impulsivity over time (all $p < .05$). Together, these findings highlight distinct developmental patterns of impulsivity across trajectory groups, and while both the youth peak and the adolescent increasing group showed elevated impulsivity, their developmental time courses diverged: the youth peak group exhibited steeper declines that were comparable to the low use group,

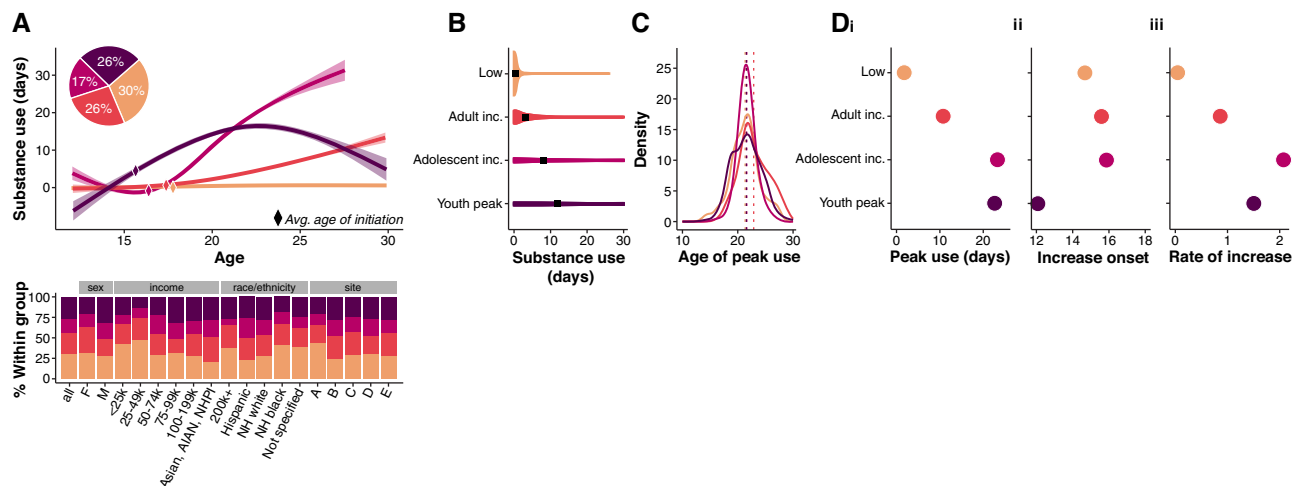


Fig. 6 | Distinct within-person trajectories in substance use. A Top panel. Growth Rate Mixture Models revealed four distinct patterns of within-person trajectories in past 30-day composite substance use, including a low group (no- or perpetually low levels of substance use across visits; $N = 241$, ~30% of participants), youth peak (earliest increases, with peak use in adolescence/young adulthood followed by decreases into adulthood; $N = 211$, ~26%), adolescent increasing (early, steep increases in use from adolescence into adulthood; $N = 139$, ~17%), and adult increasing (low use in adolescence followed by linear increases in use into adulthood; $N = 211$, ~26%). The main plot shows developmental trajectories for composite substance use across all visits. Each line represents an individual trajectory group. Fit lines are from GAMM models (via *mgcv*⁶⁴, all $s(\text{age}) p < .001$ assessed via approximate F-tests). Average age of initiation across individual trajectory groups is represented by diamonds. The pie chart indicates the percentage of participants within each trajectory group, with color corresponding to trajectories in the main plot. **Bottom panel:** Membership within each trajectory group across population

subgroups, including 18 subgroups comprising race/ethnicity, household income, guardian education, sex, and study site (geographical location). Y-axis reflects the proportion of each trajectory group across each demographic factor (shown on the x-axis). **B** Violin plots show distribution of self-reported past 30-day composite substance use across individual trajectory groups. Black squares depict mean use. Color in (B) defines the key for each trajectory group in (A-D). **C** Histograms show age of peak use across individual trajectory groups. Mean peak age is depicted by hashed lines. **D** Circles represent **i.** Average peak use across trajectory groups (mean of per-participant peak value reflecting the maximum value across all timepoints). **ii.** Average age of increase onset across trajectory groups (first age where confidence interval of the derivative did not include zero). **iii.** Rate of increase across trajectory groups (calculated as the average first derivative of the age spline). Average (Avg.). Increasing (Inc.). Female (F). Male (M). American Indian and Alaska Native (AIAN). Native Hawaiian and Pacific Islander (NHPI). Non-Hispanic (NH). Increasing (Inc.). Source data are provided as source data file 'Fig. 6'.

whereas the adolescent increasing group showed persistently elevated impulsivity into adulthood. Importantly, these results were robust to adjustment for early SU exposure (prior to age 14, see methods, see Tables S12A – S13A).

Distinct developmental patterns of inhibitory control across trajectory groups

Mean anti-saccade performance did not differ across trajectory groups (Fig. 7F, Gi; $F = 1.52$, $p = .21$, $\text{partial } R^2 = .009$, model predicted difference of ~4 points, 6% of observed range in performance), however, the youth peak group tended to have lower inhibitory control performance compared to the low, adolescent increasing, and adult increasing groups. Age-related increases were present across all groups (Fig. 7E; p s ranging from .01 to $<.001$), however, qualitatively trajectories differed in shape (Fig. 7E), change onsets, age of maturation (Fig. 7Gii), and rate of change (Fig. 7Giii). The low and adult increasing groups showed steep, rapid, age-related improvements throughout adolescence followed by stabilization in young adulthood. In contrast, age-related gains in the adolescent increasing group continued through approximately 17 years of age and qualitatively declined thereafter, although this decrease did not reach statistical significance. Finally, the youth peak group showed slower and overall attenuated age-related increases compared to other groups. Note that the anti-saccade task was administered at two participating research sites, resulting in a reduced sample size for trajectory analyses (See Methods, see Tables S8C–S11C).

We did not observe a significant trajectory group by age interaction on inhibitory control ($F = 1.13$, $p = .32$), and post-hoc tests revealed no significant differences in predicted anti-saccade performance in among groups in early adolescence (estimates for age 14, see methods,

Fig. 7H top panel, all $p > .05$), or late adolescence (estimates for age 23, see methods, Fig. 7H top panel, all $p > .05$). There were no significant differences in slopes between trajectory groups (Fig. 7H bottom panel). Together, these findings highlight statistically comparable developmental patterns of inhibitory control across trajectory groups, despite qualitative differences (see Tables S12B – S13B).

Distinct developmental patterns of NAcc tissue iron across trajectory groups

Differences in $nT2^*w$ across trajectories, as well as interactions with age, were most prominent in the NAcc and are therefore presented here and in Fig. 7. NAcc $nT2^*w$ differed across trajectory groups (Fig. 7I, Ki; $F = 3.07$, $p = .03$, $\text{partial } R^2 = .007$, model predicted difference of ~.001 units, 3% of observed range in $nT2^*w$), with higher NAcc $nT2^*w$ (indicating lower iron) in the youth peak group, followed by the low group, the adolescent increasing group, and finally, the adult increasing group. Age-related decreases were evident across all groups (Fig. 7I; $p < .001$ in all 4 models), however, trajectories differed in shape (Fig. 7I) and rate of change (Fig. 7Kiii). The low and adult increasing groups showed consistent, yet shallow, linear age-related decreases over time (indicating increases in tissue iron), while the youth peak and adolescent increasing groups showed curvilinear trajectories, with steeper, early age-related changes. Notably, the youth peak group, showed more prolonged and more rapid age-related changes in comparison to the adolescent increasing group (see Tables S8D–S11D).

We observed a significant trajectory group by age interaction on NAcc $nT2^*w$ ($F = 3.32$, $p = .007$), and post-hoc tests revealed that in early adolescence (estimates for age 14, see methods, Fig. 7L top panel), prior to the mean age of SU initiation, the youth peak group showed significantly higher $nT2^*w$, indicating lower tissue iron,

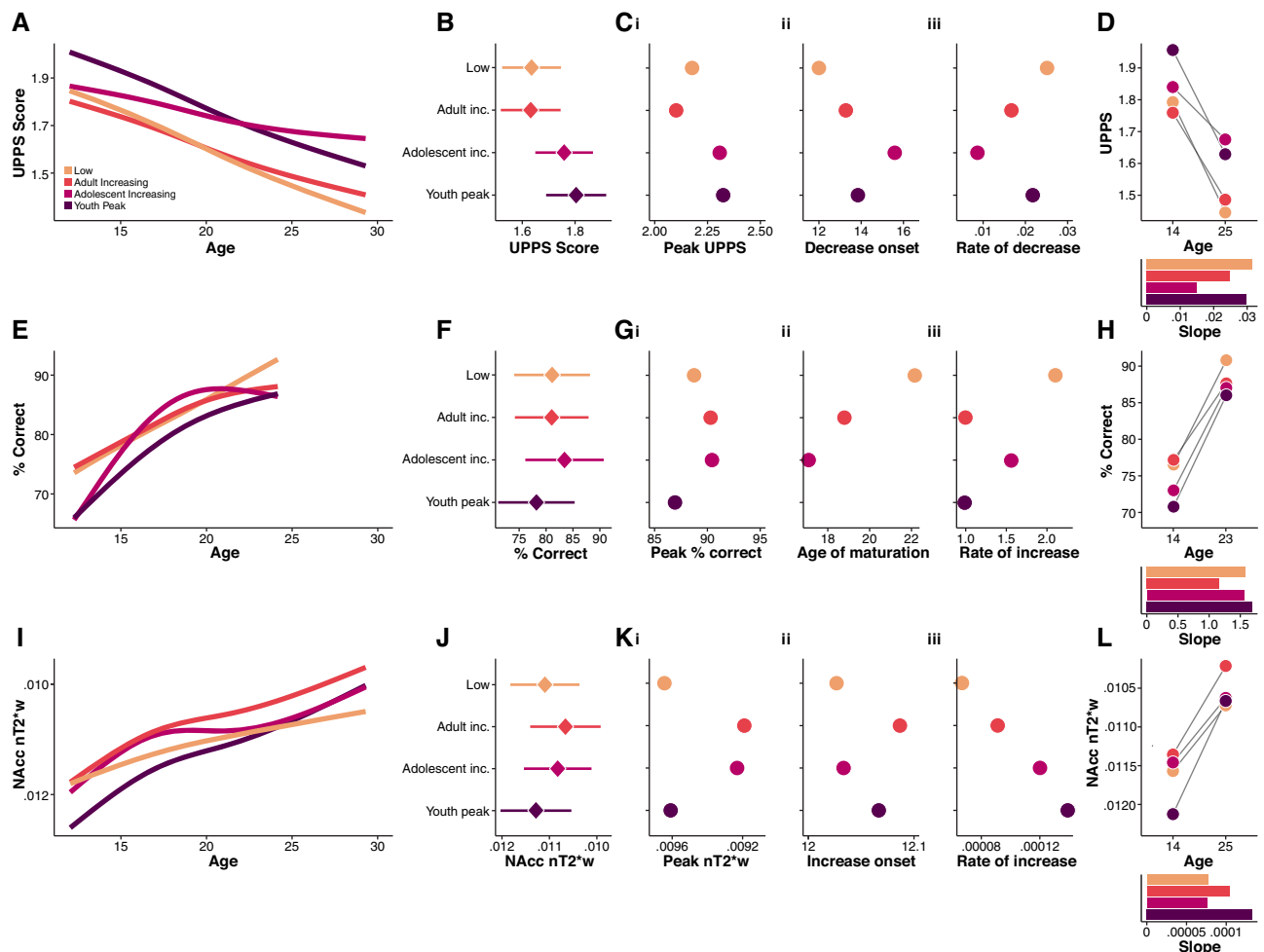


Fig. 7 | Variation in impulsivity, inhibitory control, and tissue iron within substance use trajectory subgroups. **A** GAMM-predicted trajectories of mean UPPS scores shown for each substance use trajectory group (via *mgcu*¹⁶⁴, all $s(\text{age})$ $p < .001$ assessed via approximate F-tests). **B** Diamonds indicate estimated marginal means ($\pm$ SEM; via *ggpredict*¹⁶⁵) within each group. **C** Circles represent **i.** Average peak (mean of per-participant maximum value across visits), **ii.** Average age of decrease onset (first age where confidence interval of the derivative did not include zero, $p < .05$, two-sided), and **iii.** Rate of decrease (calculated as the average first derivative of the age spline) in UPPS scores. **D Top:** Estimated marginal means for ages 14 (i.e., lower age bound) and 25 (i.e., upper age bound) across groups. **Bottom:** Corresponding slopes across age, computed from GAMM-derived marginal predictions. **E** Trajectories of mean anti-saccade performance (all $s(\text{age})$ $p < .01$). **F** Diamonds indicate estimated marginal means ($\pm$ SEM) within each group. **G** Circles represent **i.** Average peak (mean of per-participant maximum value across visits), **ii.** Average age of maturation (first age where confidence interval of the derivative included zero), and **iii.** Rate of increase (average first derivative of the age

spline) in performance. **H Top:** Estimated marginal means at ages 14 and 23 (due to restricted age range) across groups. **Bottom:** Corresponding slopes across age, computed from GAMM-derived marginal predictions. **I** Trajectories of NAcc nT2*w (all $s(\text{age})$ $p < .001$). **J** Diamonds indicate estimated marginal means ($\pm$ SEM) within each group. **K** Circles represent **i.** Average minimum value (mean of per-participant minimum value, reflecting peak tissue iron, across visits), **ii.** Average age of decrease onset, reflecting tissue iron increase onset (first age where confidence interval of the derivative did not include zero), and **iii.** Rate of decrease, reflecting increase in tissue iron (average first derivative of the age spline). **L Top:** Estimated marginal means for nT2*w at ages 14 and 25 across groups. **Bottom:** Corresponding slopes across age, computed from GAMM-derived marginal predictions. Y-axis in (I), the top panel of (L), and x-axes in (J), (Ki), and (Kii) are reversed such that visually higher values reflect higher tissue iron (lower nT2*w). Increasing (Inc). Time-averaged and normalized T2*w (nT2*w). Source data are provided as source data file 'Fig. 7'.

relative to all other trajectory groups (all $p < .05$), however, this difference was diminished by young adulthood (estimates for age 25, see methods, Fig. 7L top panel, all $p > .05$). Additionally, the youth peak group showed significantly higher slopes in comparison to the low group ($p = .006$, Fig. 7L bottom panel). Together, these findings highlight distinct developmental patterns of tissue iron across trajectory groups, suggesting that while the youth peak showed significantly lower tissue iron early in adolescence, they also exhibited steeper age-related increases, such that group differences were no longer evident in adulthood. This pattern may reflect delayed NAcc tissue iron development followed by a period of accelerated accumulation that leads to convergence with adult iron levels. Importantly, these results were robust to adjustments for early SU exposure (prior to the age of

14, see methods), and were additionally present at the level of the whole basal ganglia ($F = 2.60$, $p = .004$, see Tables S12C – S13C).

Together, these results suggest that differential substance use trajectories are associated with distinct developmental profiles of impulsivity and DA-related tissue iron – most prominently early in adolescence.

Discussion

This study examined how tissue iron, reflecting dopamine (DA) – related neurobiology, and neurocognitive functioning relate to substance use patterns from adolescence into adulthood. By integrating longitudinal MR-based estimates of basal ganglia tissue iron, an indirect correlate of DA neurobiology including presynaptic DA

availability^{69,70} with repeated assessments of impulsivity, inhibitory control, and substance use, we identified key biological and cognitive factors associated with the frequency and developmental time course of youth substance use. Across a large heterogeneous cohort and high-risk sample⁸⁰, elevated impulsivity, reduced inhibitory control, and lower tissue iron were independently associated with greater substance use frequency and polysubstance use engagement, and with distinct developmental trajectories. Growth mixture models revealed four trajectory groups, including no- or low use, youth peaks, adolescent increasing, and adult increasing, highlighting meaningful heterogeneity in developmental patterns. The youth peak and adolescent increasing groups showed elevated impulsivity early in adolescence, and the youth peak group additionally exhibited lower tissue iron, suggesting that this neurobehavioral profile may reflect a predisposition toward earlier engagement in substance use. Importantly, divergence in long-term use patterns across trajectory groups were further supported by unique maturational changes in impulsivity and tissue iron. Collectively, these results suggest that biological changes within brain systems that are still undergoing development in adolescence, including DAergic and prefrontal-dependent cognitive systems, may contribute to youth patterns of substance use. This study uniquely links longitudinal changes in basal ganglia tissue iron and neurocognitive control with substance use trajectories across adolescence, prior to the escalation of substance use, advancing our understanding of neurodevelopmental processes underlying substance use propensity in youth.

Across the full sample, substance use showed robust non-linear developmental pattern, increasing through adolescence into young adulthood, followed by either stabilization or declines toward the upper limit of the study period. Given that a primary objective of the current study was to identify neurodevelopmental differences that were predictive of variation in substance use trajectories (i.e., adolescent-specific vs long-term escalation into adulthood), we utilized an innovative growth mixture modelling approach that identified subgroups based on patterns in within-person trajectories of substance use across time points. Trajectories clustered into four distinct subtypes across the population (Fig. 6A). The low-use group (~1/3 of participants) showed minimal use across the study period and later initiation, consistent with recent statistics showing that fewer youth are using alcohol⁸⁵. The youth peak group (~1/4 of participants) showed the earliest initiation and highest overall use, with quadratic patterns across adolescence marked by rapid early increases, peaks in use in adolescence/young adulthood, notably followed by declines beginning around the mid-twenties, potentially reflecting normative adolescent rises in risk-taking propensity^{86,87}. The adolescent increasing group (~1/5 of participants) showed early initiation, and the second highest rates of substance use overall, with steep linear increases beginning in mid-adolescence, potentially indexing higher risk for sustained patterns of substance use into adulthood. Finally, the adult increasing group (~1/4 of participants) showed low use during adolescence, later initiation and slower rates of change, with young adulthood peaks in use and continued increases into adulthood (toward the end of the study period). Trajectory patterns in magnitude and timing generalized across substances, reflecting distinct pathways of substance use across adolescence into adulthood, and establishing a foundation for testing how neurocognitive and neurobiological maturation shape divergence in trajectories across adolescence.

Impulsivity relates to developmental trajectories of substance use

Across the full sample, impulsivity was highest early in adolescence and showed robust linear decreases until the mid-twenties across subscales. Sensation seeking showing relatively prolonged elevation with attenuated age-related decreases beginning at approximately 18 years of age, consistent with prior reports of adolescent sensation

seeking^{88,89} and risk-taking^{17,86}. Increased impulsivity emerged as a robust and consistent marker of substance use magnitude (days of use), frequency, and polysubstance use, and further predicted variation in trajectories. These findings were similar across substances and subscales, supporting a broad role for impulsivity in shaping youth patterns of substance use. Specifically, youth who followed early-onset trajectories (youth peak and adolescent increasing) showed elevated impulsivity early in adolescence – often preceding the mean age of substance use initiation (and remaining robust after adjusting for early substance exposure). However, the developmental time course distinguished trajectory groups: individuals in the youth peak group exhibited steeper age-related declines in impulsivity, converging toward the low-use group in young adulthood, concomitant with decreases in substance use. In contrast, those in the adolescent increasing group showed persistently elevated impulsivity and attenuated developmental decreases. This divergence may suggest that while early heightened impulsivity early in adolescence increases vulnerability for early initiation, its developmental time course may underlie whether subsequent use declines into adulthood or escalates. Thus, normative maturation of impulsivity may facilitate reductions in risk-taking for some youth, whereas sustained impulsivity may contribute to prolonged vulnerability.

Inhibitory control relates to developmental trajectories of substance use

Inhibitory control robustly improved across adolescence and stabilized in young adulthood, consistent with prior anti-saccade studies^{33,90–92} and executive function maturation more broadly^{92,93}. Overall, lower inhibitory control was moderately associated with greater substance use, particularly in terms of frequency and polysubstance use engagement. Associations were comparable across trial types, suggesting generalized inhibitory control involvement rather than a reward-specific mechanism, and are complementary to prior reports in NCANDA-A showing decreased inhibitory control performance in youth at increased risk for substance use^{4,15}, most prominently for overall substance use, cannabis, and nicotine. Differences across trajectory groups were not statistically robust, potentially due to decreased power as anti-saccade task performance was assessed only at two study sites. However, qualitatively, the youth peak trended toward lower performance and smaller age-related gains, while the adolescent increasing group showed declines in performance toward late adolescence, potentially suggesting delayed or attenuated inhibitory control maturation that could contribute to elevated levels of use.

Dopamine neurobiology relates to developmental trajectories of substance use

Consistent with prior EPI-derived tissue iron studies^{69,72–75,94}, nT2*_w decreased across basal ganglia regions, reflecting age-related increases in tissue iron that parallel animal findings on DA availability during analogous periods^{30,95,96}. Across the full sample, lower basal ganglia tissue iron was moderately associated with increased substance use magnitude and frequency, particularly overall substance use, alcohol use, binge drinking, and cannabis, and further differentiated substance use trajectories. Trajectory-related differences were most apparent in the nucleus accumbens (NAcc), with youth peak trajectories associated with moderately lower tissue iron, most prominently early in adolescence, preceding the mean age of initiation (a finding that was robust to adjustment for early substance use exposure). Interestingly, relative to other basal ganglia ROIs (e.g., the dorsal striatum), the NAcc exhibited earlier and more rapid increases, with peak change occurring at approximately 13 years. Given links between tissue iron and DA function^{62,69,73,97} this earlier NAcc pattern may be mechanistically related to adolescent peaks in reward reactivity^{12,29,33,98,99} and limbic

influence^{100,101}, contributing to increased substance use and potentially, earlier initiation. These findings are consistent with ‘inverted-U’ hypotheses of DA function^{102,103} and neurodevelopmental models suggesting that low DA signaling might promote compensatory risk-taking and reward seeking behavior^{27,104}. In light of the current findings, it is possible that low tissue iron may reflect low basal DA, or blunted endogenous DA, and promotes increased motivational salience of external rewards, including substances, as a means of acutely boosting DA signaling^{105–107} and providing hedonic stimulation¹⁰⁸, particularly in youth who show early substance use engagement. Importantly, despite early differences, the youth peak group also showed the steepest age-related increases in tissue iron, such that differences in nT2*w were not evident later in young adulthood. One interpretation is that low tissue iron, reflecting low DA levels, in early adolescence may predispose youth to earlier and more frequent substance use, but compensatory accelerated iron accumulation (the observed “catch up”) may support later reductions in use into adulthood. Trajectory-based effects were most pronounced in the NAcc, aligning with its well-established role in reward-, risk-taking^{12,29,32,33,109}, and substance use more broadly^{18,19,78}, while more dorsal striatal regions may be more relevant for later maintenance and habitual aspects of substance use¹¹⁰. However, similar patterns were evident at the level of the basal ganglia, and the present data do not allow definitive conclusions about NAcc-specific mechanisms.

A central contribution of this study is the demonstration that variation in DA-related neurobiology and neurocognitive systems is detectable early in adolescence and relates to distinct developmental pathways of substance use. These findings suggest that pre-existing differences in developing reward and control systems may bias trajectories even before substantial use occurs, distinct from long-term consequences of substance use exposure. Importantly, impulsivity, inhibitory control, and tissue iron each independently related to substance use, indicating distinct contributions of reward sensitivity, cognitive control, and DA-related neurophysiology. Together, they support a developmental model in which the maturation of DA-sensitive mesocorticolimbic systems and prefrontal control systems jointly shape adolescent risk-taking behavior.

Sociodemographic generalizability

Although overall levels of substance use, neurocognitive, and neurobiological makers varied across sociodemographic factors, the developmental timing and shape of trajectories were largely conserved. Age of initiation, peak periods, demarcated periods of change, and rates of change across measures were broadly consistent across subgroups (see Supplemental Results for full description), and no sociodemographic group was disproportionately represented within a given trajectory group. These findings suggest that developmental windows of risk and their neurobehavioral mechanisms may be broadly generalizable across diverse populations, aligning with evidence that the timing of adolescent risk-taking is relatively conserved across social, environmental, and psychological factors⁸⁶. Although prior research has documented variability in the magnitude of substance use and neurocognitive functioning across demographic groups, far less attention has been paid to the developmental timing. This distinction is crucial for advancing mechanistic models of the etiology and pathogenesis of substance use disorders, which often emerge during adolescence, and for informing the design of developmentally sensitive interventions. Furthermore, while previous studies have reported age-related changes in tissue iron and neurocognitive metrics, the precise timing, magnitude, and periods of maturation have remained poorly defined. Leveraging the large and deeply phenotyped sample from NCANDA-A, we were able to delineate specific developmental windows during which substance use behaviors and their associated biological underpinnings undergo significant change.

Strengths and implications

A key strength is the longitudinal design, enabling the characterization of within-person changes over time, which is essential for disentangling predisposing neurobiological factors from long-term consequences of substance use. The primarily substance-naïve baseline (83% limited/no history of use) allowed us to capture early markers that may prospectively predict substance use frequency and trajectory. The wide age range (12–30 years) additionally enabled differentiation in long-term use patterns – allowing us to distinguish transient, potentially normative increases in use during adolescence and young adulthood (e.g., sensation seeking) from more persistent escalation and may reflect heightened risk. By analyzing both a composite substance use measure as well as substance-specific effects (including nicotine, cannabis, and alcohol), we captured underlying DA-related and neurocognitive patterns linked general substance use propensity, while also retaining specificity. From a translational perspective, these findings highlight early adolescence as a potentially sensitive window for intervention. Identifying youth with elevated impulsivity and lower DA-related maturation prior to substance use escalation may allow targeting of prevention strategies. More broadly, this work bridges animal and human literature by linking DA-related processes to patterns of substance use across development.

Opportunities for ongoing investigations and methodological limitations

Several limitations highlight opportunities for important future research. First, we did not test whether trajectory group membership predicts long-term clinical outcomes, including substance use disorders or neuropsychiatric conditions. Future work should determine whether these developmental trajectories differentially forecast adult clinical risk. Second, the adult increasing group did not show clear differences in neurocognitive or neurobiological factors relative to low use. This pattern suggests that that later-onset increases may be driven by factors not captured here (e.g., contextual or environmental stressors), warranting investigation into mechanisms underlying adult-onset substance use. Third, between-person analyses cannot fully disentangle whether characteristics identified here precede substance use or are partially shaped by it. However, trajectory analyses anchored in early adolescence (age 14), prior to mean age of initiation and adjusted for early exposure, suggest that early group differences are unlikely to reflect substance-induced effects. Relatedly, identifying contributions of individual differences prior to adolescence remains critical. Genetic and epigenetic variation in DA-related genes^{111,112}, prenatal exposures^{113,114}, and early environmental factors^{115–118} have been linked to alterations in DA-sensitive circuit development and may shape vulnerability to later substance use.

Fourth, future work should integrate contextual and environmental factors known to shape substance use patterns, including peer norms, parental/societal attitudes^{1,119}, and youths’ motivations (e.g., coping with stress or boredom^{120–122}) as potential drivers or distinguishing factors in substance use behavior. Pubertal timing and associated hormonal changes may further contribute to variability in substance use trajectories¹²³ by altering sensitivity to reward, peer influence, and health-related decision-making. Finally, broader contextual factors, including changing policies and legislation (i.e., smoke-free laws for tobacco use¹²⁴ or the legalization of cannabis in various countries and states¹), as well as societal disruptions during the COVID-19 pandemic¹, may influence youth engagement with substances. Large-scale longitudinal studies and ongoing surveillance efforts, such as the such as the Adolescent Brain and Cognitive Development (ABCD) study¹²⁵ and the Monitoring the Future study¹ will be critical for understanding biological risk and shifts in youth substance use over time.

Methodologically, T2*-based indices provide a non-invasive but indirect proxy for tissue iron and DA-related neurobiology, but the T2* signal can also be influenced by other neurodevelopmental processes, including myelin^{126,127}. Therefore, analyses focused on the basal ganglia, where iron concentration most reliably drives T2* variation^{47,48,128}. Future studies incorporating quantitative MRI approaches such as R2*, R2', or Quantitative Susceptibility Mapping (QSM)^{69,74} may improve specificity and sensitivity, and potentially enable examination of DA-related processes in broader mesocorticolimbic circuits, including PFC. However, EPI-derived iron measures show strong correspondence with more direct measures including R2'¹²⁹ and susceptibility-based indices in midbrain regions ($r_s = .62-.84$)¹³⁰ and similar trajectories⁷⁴, supporting their validity as developmental markers of basal ganglia tissue iron. Additionally, although nT2*w was derived from EPI data with relatively coarse native voxel size – raising concerns about potential partial-volume effects that may contribute to higher between-person variability in smaller structures – NAcc distributions and elevated interindividual variability shown here match prior studies across a range of modalities^{17,69}, resolutions^{72,75}, and field strengths⁹⁴, including quantitative relaxometry-based measures, suggesting this reflects a biological property of this region rather than solely measurement factors. Nevertheless, we emphasize results at the level of the basal ganglia as a whole and subregional specificity warrants future investigation.

Finally, indices of inhibitory control were available in a reduced subset ($n = 267$; two sites; visits 1 – 5), limiting power for subgroup analyses and longitudinal coverage. Supplemental analyses restricting impulsivity and tissue iron analyses to this subset preserved overall developmental patterns, trajectory group composition, and robust impulsivity-substance use associations, with tissue iron-substance use associations attenuated but directionally consistent. The truncated age-range (≤ 24 years) likely reduced sensitivity to later-emerging divergence across trajectories, likely contributing to weaker effects for inhibitory control. Accordingly, inhibitory control findings should be interpreted cautiously, while the within-person trajectory framework and core associations across domains remain robust.

Developmental implications

Adolescence is marked by dynamic maturation of DA-related and cognitive control systems. This study demonstrates that developmental variation in these systems relates to how substance use emerges and unfolds across adolescence into adulthood. Elevated impulsivity, diminished inhibitory control, and lower DA-related tissue iron characterize youth with greater substance use engagement, and moreover, maturational profiles in these systems distinguished distinct developmental trajectories of substance use. Differences emerged early in adolescence, prior to mean substance use initiation, and for some individuals, appeared to stabilize over time, potentially through age-related maturation of DA systems, normative decreases in impulsivity, and improvements in cognitive control. In others, attenuated age-related decreases that lead to persistent impulsivity may contribute to prolonged substance use. Together, these findings support neurodevelopmental models in which the maturation of mesocorticolimbic DA-sensitive systems and cognitive control systems shapes vulnerability to adolescent substance use, and emphasize early adolescence as a critical window for identifying and mitigating risk before problematic use emerges.

Methods

Participants

Data were provided from the National Consortium on Alcohol & Neurodevelopment in Adolescence and Adulthood (NCANDA-A) study⁸⁰ at five sites across the United States: University of Pittsburgh Medical Center (UPMC), SRI International (SRI), Duke University (Duke), Oregon Health & Science University (OHSU), and University of

California San Diego (UCSD). All experimental procedures obtained full Institutional Review Board (IRB) approval at each site, and complied with the Code of Ethics of the World Medical Association (Declaration of Helsinki, 1964)¹³¹. Eight hundred and thirty one adolescents and young adults (423 female; age range = 12–22 years old at baseline) were enrolled in an accelerated longitudinal study (total participant visits = 6271; median number of visits per-participant = 5; range of visits per-participant = 1–9; age range = 12–30 years old across all longitudinal visits). Participants were screened for study exclusion criteria including MRI contraindications (e.g., claustrophobia, non-removable metal in the body), head injury with significant loss of consciousness, psychiatric disorders that might influence study completion (e.g., psychosis), and psychiatric medication⁸⁰. At each visit, participants completed a neuropsychological battery, 3 T magnetic resonance imaging (MRI) neuroimaging session, and a comprehensive assessment of substance use, psychiatric symptoms, and diagnoses as well as functioning in major life domains⁸⁰. Given that a central goal of NCANDA-A was to examine the transition to significant substance use during adolescence into adulthood, approximately 50% of participants were recruited based on subclinical risk factors for alcohol use disorder, however, included participants had limited or no history of alcohol or other drug use at baseline as per National Institute on Alcohol Abuse and Alcoholism (NIAAA) guidelines for risky drinking⁸⁰. Following these exclusion criteria, the final sample included eight hundred and two participants (398 female; age range = 12–22 years old at baseline, total participant visits = 6078; median number of visits per-participant = 5; range of visits per-participant = 1–9; age range = 12–30 years old across all included longitudinal visits). To optimize generalizability and representation (see^{86,132,133} for discussion) and due to studies demonstrating that the developmental timing of adolescent risk-taking behavior is consistent across sociodemographic factors⁸⁶, no participant-level demographic exclusion criteria were applied to the analysis sample, and we additionally show the potential effect of such factors in a series of sensitivity analyses. Participants over 18 provided informed consent, while participants younger than 18 provided written assent and parental consent, and participants were compensated for completing research assessments. Each site recruited representative community samples with respect to local racial/ethnic distributions, ensuring balance for equal distribution of sex by age group. Participant demographic information is provided in Table 1 (but see⁸⁰ for detailed sampling strategy, recruitment information, and assessment protocol).

Substance use assessments

At each visit, participants completed the self-report Customary Drinking and Drug-use Record (CDDR), which assesses the frequency (never, ever, regular (weekly) use) and quantity of recent (i.e., number of days in the past 30 days) alcohol and substance use^{80,134}. We focused our analyses on alcohol (including binge drinking), nicotine, and cannabis use. Binge drinking was assessed based on the NIAAA guidelines of consuming greater than four or five drinks for females and males, respectively, within an occasion. Age at first use and age of initiation of were also recorded. Data (frequency, use-type, and quantity) were coded separately for each substance type at each visit.

Impulsivity measures: UPPS-P

Impulsivity was assessed at each visit using the self-report Short UPPS-P Impulsive Behavior Scale (SUPPS-P)^{81–83}, which includes subscales assessing five distinct dimensions of impulsivity, including sensation seeking, negative urgency, positive urgency, perseverance (lack of), and premeditation (lack of). Each subscale ranged from 1 to 4, with 4 being the maximum score. Primary analyses used the mean across subscales to index global trait-level impulsivity, with subscale-specific results reported in the supplement.

Executive function measures: anti-saccade task

Two hundred and seventy-six participants (150 female, 12–24 years old, 1–5 visits, 682 total sessions) at two participating sites (Duke & UPMC) additionally completed a rewarded anti-saccade task (Fig. 3A), as has been previously been described^{15,33,84} and in the Supplement. The full protocol included a total of 56 full incentivized ('reward') and 56 full non-incentivized ('neutral') anti-saccade trials, completed across 4 fMRI runs, each consisting of 28 trials for a total of 112 trials. Briefly, the anti-saccade task is a visuospatial task in which participants are presented with a peripheral cue and required to direct their gaze to its mirror location, requiring the suppression of the reflexive response to look toward the stimulus and the generation of a voluntary eye movement in the opposite direction. See Supplemental Methods for the full task description including stimulus presentation, timing, eye tracking recording and apparatus.

The behavioral indicator of interest was accuracy (the percentage of correct anti-saccade responses) which was calculated as the number of correct trials divided by the number of scorable (tracked) trials and was calculated separately for reward and neutral trials. Sessions with fewer than 25 viable trials total, as well as correct response rates lower than 40%, were excluded from analysis, resulting in a final sample of 257 12–24-year-old participants (134 female), for a total of 586 sessions.

MRI data processing

Structural imaging acquisition & preprocessing. Multisite protocols ensured that comparable 3 T MR data were acquired across the 5 imaging sites (methods have been reported in detail^{15,135}). Two sites used Siemens TIM-Trio scanners (UPMC and OHSU) and three sites used GE Discovery MR750 scanners (UCSD, SRI International, & Duke); however, all participants underwent the similar scan protocols at each timepoint. Structural (T1-weighted) images were acquired on Siemens scanners using a magnetization-prepared rapid gradient-echo (MPRAGE) pulse sequence (TR, 6 ms; TE, 3 ms; flip angle, 9°; matrix, 256 × 256; FOV, 24 cm; 1.2 × .94 × .94 mm voxel size, 160 slices), and on GE scanners using an Inversion Recovery-Spoiled Gradient Recalled echo (IR-SPGR) sequence (TR, 6 ms; TE, 2 ms; flip angle, 11°; matrix, 256 × 256; FOV, 24 cm; 1.2 × .94 × .94 mm voxel size, 146 slices). Structural data were preprocessed to extract the brain from the skull, and warped to the MNI standard brain using both linear (FLIRT) and non-linear (FNIRT) transformations.

Acquisition & preprocessing of T2*-weighted indices reflecting tissue iron properties

T2*-weighted MR data were acquired over a 10-minute fixation resting state scan using a 3D blood oxygen level dependent (BOLD) echoplanar sequence (TR, 2,200 ms; TE, 30 ms; flip angle, 79°; 3.75 × 3.75 × 5 mm voxel size; 32 slices; 275 volumes) providing maximal coverage of the cortex and basal ganglia. T2* data were preprocessed using in-house analysis scripts¹³⁶ (10.5281/zenodo.8320245) including simultaneous slice-timing and motion correction using NiPype's SpaceTimeRealign. Data were minimally preprocessed as required for our T2*-based analyses, and included 4D slice-timing and head motion correction, skull stripping, co-registration to the participant's structural image, and nonlinear warping to 3-mm isotropic resolution MNI space (matrix size = 65 × 77 × 65)^{73,75}. The average proportion of high-motion volumes across participants was 6.3%. Because we used an FD threshold to censor individual volumes, no participants were excluded for high motion (see below).

Time-averaging and normalization of T2 data (nT2*w). T2*-weighted BOLD images were preprocessed as in^{73–75,137,138}, including the following steps: Each volume was first normalized to the whole-brain mean (z-score normalization) using a coverage map created from all inputs for each participant including only non-zero values. This normalized signal was then aggregated voxel-wise across all volumes in the resting

state scan using the median, which reduces the effect of outlier volumes resulting in one normalized T2*-weighted image for each participant (nT2*w). Volumes containing frame-wise displacement (FD) > 0.3 mm were identified as high motion timepoints and were excluded from analyses¹³⁹. This procedure ensures enhanced SNR by averaging across multiple timepoints⁷³, and the normalization provides the T2* decay in the basal ganglia relative to the whole brain, which allows for comparison of nT2*w values across participants. Critically, as shorter T2* relaxation times indicate stronger magnetic field inhomogeneities induced by iron, nT2*w is inversely related to iron, with lower values indicating *higher* iron content, and higher values indicating *lower* iron content.

Regions of interest. Given evidence from prior studies showing mixed results in terms of specificity across basal ganglia ROIs in modulation of cognitive and reward function during adolescence⁷⁵, we extracted nT2*w values across all basal ganglia regions of interest defined using the Harvard-Oxford subcortical atlas¹⁴⁰ (Fig. 4A) including the globus pallidus (GP, 62 voxels), nucleus accumbens (NAcc, 18 voxels), putamen (183 voxels), and caudate nucleus (111 voxels), and additionally computed one per-participant estimate of nT2*w that reflected the mean across all voxels in the basal ganglia (374 voxels). ROI masks were applied using a 25% probability threshold, and estimates were computed as the mean nT2*w signal across voxels within each bilateral ROI. Because nT2*w was derived from EPI data with relatively coarse native voxel size, estimates in small structures such as the NAcc may be more susceptible to partial-volume effects. Accordingly, NAcc findings are interpreted at the level of the anatomical ROI rather than fine-grained subregional localization, and we additionally interpret the majority of prior findings at the level of the whole basal ganglia. Consistent with prior work^{75,137}, basal ganglia estimates of nT2* were longitudinally stable across sessions, evidenced by high intra class correlation coefficient (ICCs, Psych Package, version 2.1.6¹⁴¹) of nT2*w values across visits (.89). nT2*w indices of basal ganglia physiology were available in 799 12–29-year-old participants (395 female) for a total of 4363 sessions.

Harmonization of nT2*w across scanners. nT2*w estimates from BOLD data approximates the T2* decay time based on the mean amplitude of BOLD signal normalized to the whole-brain amplitude, which could result in variation in absolute values for nT2*w across different scanners, software versions, and scan parameters. Given the multi-site nature of NCANDA-A, as well as variation in scanner manufacturer (GE, Siemens) across sites, and changes to scanner model version within sites over time, we harmonized nT2*w data using NeuroCombat (*neuroCombat* package version 1.0.13 in R¹⁴²). We found that harmonizing based on scanner model version ameliorated substantial variability in nT2*w values relative to harmonizing based on site or scanner manufacturer alone, and additionally improved the consistency of age-effects and minimized between-site variation in nT2*w. Thus, all nT2*w values reflect adjusted estimates based on harmonization procedures.

Data analysis

Characterization of substance use patterns. Participants' responses from the CDDR were assessed at each visit in the following ways for past 30-day substance use: (1) Number of days of use within the past 30 days [alcohol, cannabis, nicotine], number of binge drinking days within the past 30 days (based on NIAAA guidelines, see above); (2) Responses were additionally grouped into the following frequency categorizations for each substance: never (reporting never having used at that timepoint), ever (reporting having ever used), and regular (reporting having used at least once/week); (3) Responses were grouped into the following use-type categorizations for each substance: no-use (reporting never having used that substance), co-use

Table 1 | Participant characteristics at baseline

Characteristic	Category	%	N
Race/Ethnicity	Asian, AIAN, NHPI	8.23	66
	Hispanic	11.97	96
	NH Black/AA	12.09	97
	NH White	63.34	508
	Unspecified	4.36	35
Household Income ^a	<25k	5.49	44
	25k-49k	10.35	83
	50k-74k	11.47	92
	75k-99k	12.22	98
	100k-199k	41.02	329
	>200k	19.45	156
Guardian Education (Average)	<Grade 9	.87	7
	High school	7.48	60
	College	49.13	394
	Graduate School	42.52	341
Sex	F	49.63	398
	M	50.37	404
Age at Baseline	12–14	31.92	256
	15–17	38.40	308
	18–22	29.68	238
Site	A	14.84	119
	B	20.07	161
	C	20.57	165
	D	18.70	150
	E	25.81	207
Scanner	GE	66.46	536
	Siemens	33.54	271

Coding for race/ethnicity, household income, guardian education, sex, age at baseline, study site, and MRI scan manufacturer. Guardian education reflects average across guardians. All variables were assessed via self-report. See ‘Participants’ for information on recruitment. Non-Hispanic (NH). African American (AA). American Indian and Alaska Native (AIAN). Native Hawaiian and Pacific Islander (NHPI).

(reporting using that substance plus another substance in the past 30 days), and poly-use (reporting using that substance plus 2 or more other substances in the past 30 days). The frequency and use-type categorizations allowed us to ascertain whether effects occurred in a dose-dependent manner that scales with the number of substances that were used concurrently.

Primary analyses investigated substance use patterns overall via a probabilistic score that estimated the number of days (out of 30) that individuals used at least one substance. Follow-up sensitivity analyses investigated effects across each substance specifically, including alcohol, binge drinking, cannabis, and nicotine in order to ascertain whether risk-makers were associated with general patterns of use or whether they reflected individualized risk for different substances that might have different etiological contributors and developmental time courses (see Supplemental Tables).

General statistical approach and theoretical models. All analyses were conducted in R via RStudio version 4.3.1 (R Core Team (2023), <https://www.r-project.org>). Two streams of analyses were implemented, the first to describe age-related changes in substance use, impulsivity, inhibitory control, and basal ganglia tissue iron, and to examine effects of the latter three metrics on concurrent substance use (termed ‘between-person analyses’ below), and the second to predict whether variation in these factors were predictive of distinct

trajectories in substance use across the study period (termed ‘within-person analyses’ below).

All statistical analyses utilized generalized additive mixed models (GAMMs; *mgcv* package in R¹⁴³, version 3.5.2), including random intercepts and random slopes estimated for each participant to account for the longitudinal aspect of the dataset. All continuous variables were modelled as smooth terms (age, impulsivity, anti-saccade, and tissue iron), and sociodemographic variables including biological sex assigned at birth, family income, and study site were coded as categorical variables (factors) and were modelled as parametric terms. We used Restricted Maximum Likelihood (REML) to fit our GAMMs with unbiased estimates of the variance components in the context of random effects¹⁴⁴. All primary models were conducted across a composite measure of substance use, mean UPPS, mean anti-saccade, and basal ganglia nT2*w, and a series of sensitivity analyses repeated these analyses separately across substance, subscale, trial type, and ROI (see Supplemental Tables).

For each GAMM, we report (1) partial R^2 for the age smooth or primary predictor to quantify variance explained beyond covariates, and (2) Δ (delta) effects expressed as the model-predicted differences in the outcome across the observed range of a predictor (or across categories), reported in raw units and as a % of the observed outcome range to aid interpretability when units are arbitrary (i.e., nT2*w, see Supplementary Tables for full statistics).

Developmental trajectories in the full sample. We first characterized age-related changes across all measures (‘outcome variable’, *Model 1*) using GAMMs, which were chosen to characterize non-linear developmental trajectories given that non-linear fits have been shown to provide a better approximation of developmental changes throughout adolescence^{91,93,145–147}. Penalized natural cubic regression splines were implemented via the *mgcv* package to characterize linear and/or non-linear age trajectories^{144,148}, and a maximum basis complexity (k) of 4 was chosen consistent with prior studies utilizing GAMMs in adolescent populations^{72,149–153} based on theoretical expectations that there are unlikely to be more than three resolvable inflection points (e.g., childhood, adolescence, adulthood stages), and to prevent overfitting the age smooth function. To assess age-ranges of significant change, we calculated the average first derivative of the age spline from the GAMM using finite differences (as were conducted in^{72,154}), and 95% confidence intervals of the derivatives were generated (*gratia* package in R¹⁵⁵). Confidence intervals of the derivative that did not include zero indicated intervals of significant change. Average derivative was computed as an estimate of the average rate of change per year, across the entire age range. To additionally quantify developmental trends in the data, we computed the mean per-participant value (mean), as well as the peak value (maximum across all timepoints) across all metrics [substance use, impulsivity, inhibitory control performance, tissue iron], and we took the age (years) of this peak value to estimate the age of peak [substance use, impulsivity, inhibitory control performance, tissue iron].

Model 1 = outcome variable $\sim s(\text{visit age}, fx = F, k = 4, bs = "cs")$
 $+ \text{sex} + \text{income} + \text{race/ethnicity} + \text{site},$

random = list (ID = ~ 1 , ID = $\sim 0 + \text{visit age}$).

Participant-level sociodemographic sensitivity analyses. Sensitivity analyses explored how age-related trajectories changed as a function of 18 subgroups defined via levels of four variables (sex, race/ethnicity, study site, and socioeconomic indicators (family income), see Table 1) that have previously been shown to be associated with adolescent risk-

taking propensity⁸⁶. Independent models assessing developmental trajectories (as above) were conducted separately across these 18 subgroups to determine generalizability of developmental models.

Characterizing relationships between substance use and impulsivity, inhibitory control, and basal ganglia tissue iron. To test whether neurocognitive/neurobiological measures varied with qualitative use patterns we modelled (1). Impulsivity (UPPS), (2). Inhibitory control (anti-saccade performance), and (3). Tissue iron (nT2*w) as outcomes predicted by use frequency and use-type categories, covarying age and sociodemographics (*Models 2 & 3*).

$$\text{Model 2} = \text{outcome variable} \sim \text{category} + s(\text{age}, fx = F, k = 4, bs = "cs") \\ + \text{sex} + \text{income} + \text{race/ethnicity} + \text{site},$$

$$\text{random} = \text{list}(ID = \sim 1, ID = \sim 0 + \text{visit age}).$$

We then modelled continuous past 30-day use as the outcome predicted by each continuous measure modeled as a smooth term, including age as a smooth covariate and sociodemographic factors. Predicted smooths used thin-plate splines ($k = 3$) as we did not expect these associations to largely deviate from linearity. Continuous measures of substance use were log transformed for these analyses (*Model 3*).

$$\text{Model 3} = \log(\text{substance use}) \sim s(\text{independent variable}, fx = F, k = 3) \\ + s(\text{visit age}, fx = F, k = 4, bs = "cs") + \text{sex} + \text{income} \\ + \text{race/ethnicity} + \text{site},$$

$$\text{random} = \text{list}(ID = \sim 1, ID = \sim 0 + \text{visit age}).$$

We evaluated specificity across UPPS subscales, anti-saccade trial types, and basal ganglia ROIs via interaction models. As interactions were not significant (UPPS: $F = .40, p = .85$, anti-saccade: $F = .19, p = .83$, nT2*w: $F = 1.99, p = .09$), primary analyses used mean estimates across these measures; however, we present results across each subscale, trial condition, and ROI in the Supplemental.

Within-person trajectory classification analyses. We characterized latent trajectory groups through growth mixture models (i.e., Latent Class Mixed Model; *LCMM* package in R^{156,157}, version 2.1.0) that captured variation in within-person trajectories (patterns of change) across timepoints and classified individuals based on patterns of substance use across development¹⁵⁸ (*Model 4*). Iterative class models were performed that tested for 1–4 separate classes of substance use trajectories, and the best fitting model was chosen based on the minimum Bayesian information criteria (BIC), Akaike information criteria (AIC), entropy, and integrated completed likelihood (ICL). Random intercepts were modelled per class, allowing each class to follow its own developmental trajectory, with participant and age as random effects. The same covariate structure as above was used, with the addition of baseline age as a between-participant variable to account for the longitudinal cohort design and to account for between- and within-participant effects, and modeling center age and a quadratic trend for center age (*Model 4*). Given that fitting mixture models can be sensitive to initial values, we utilized a gridsearch function to improve and stabilize estimations of the models, and to improve the chance of finding the global maximum likelihood^{156,157}. The model was estimated 100 times with different random starting values, with each repetition running up to 30 iterations. We limited our models to 1–4 classes based on theoretical expectations of developmental heterogeneity¹⁵⁹, recommendations from prior mixture modeling research, and to ensure adequate class sizes, interpretability, and model

parsimony^{160,161}.

$$\text{Model 4} = \text{hlme}(\log(\text{substance use}) \sim \text{center age} + I(\text{center age}^2) \\ + \text{center age at baseline} + \text{sex} + \text{income} + \text{race/ethnicity} + \text{site},$$

$$\text{random} = \sim 1, \text{subject} = 'ID', ng = 4, \text{mixture} = \sim \text{center age} + II(\text{center age}^2).$$

Participant-level sociodemographic sensitivity analyses. To confirm that sociodemographic factors were not disproportionately represented within each trajectory group, sensitivity analyses explored how group membership varied as a function of the 18 sociodemographic subgroups defined above.

Characterizing whether variation in impulsivity, inhibitory control, and tissue iron contributed to within-person trajectories in substance use. Utilizing GAMMs (as above), we then tested for differences in 1. Impulsivity (UPPS), 2. Inhibitory control (anti-saccade performance), and 3. Basal ganglia tissue iron (nT2*w) across trajectory groups (*outcome variable; Model 2*).

Characterizing significant trajectory group by age interactions on substance use. We then tested for interactions with age and trajectory group on each variable that might suggest that the effect of these factors varied across development (i.e., to ascertain whether trajectory groups had differences in DA early in development prior to the onset of substance use, or in young adulthood following substance use exposure; *Model 5*).

$$\text{Model 5} = \text{outcome variable} \sim s(\text{age}, fx = F, k = 4, bs = "cs") \\ + \text{sex} + \text{income} + \text{race/ethnicity} + \text{site} + \text{trajectory group} \\ + s(\text{age}, \text{trajectory group}, fx = F, k = 4, bs = "sz"),$$

$$\text{random} = \text{list}(ID = \sim 1, ID = \sim 0 + \text{visit age}).$$

To characterize whether differences in maturational timing of impulsivity, inhibitory control, and tissue iron contributed to substance use trajectories, we repeated the growth rate modelling above within each trajectory group (*Model 1*). We further characterized significant age interactions within each trajectory group using the *ggpredict* function in R (*ggeffects*, version 1.1.5¹⁶²) by specifying fixed values for the age that approximated the range of our data (age 14 for early adolescence; age 25 for young adulthood, with the exception of interactions involving the anti-saccade task in which we tested age 23 for young adulthood due to the restricted age-range). We then plotted the predicted relationship between age and dependent variable for each trajectory group and tested for differences at each age using the *hypothesis test* function (*ggeffects*). Furthermore, we assessed differences between slopes across each measure within trajectories. To evaluate whether neurocognitive and neurobiological differences preceded substance use initiation, additional sensitivity analyses controlled for early substance use exposure (prior to age 14).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The present work used publicly available developmental and substance use data from the National Consortium on Alcohol & Neurodevelopment in Adolescence and Adulthood (NCANDA-A) study, available via the NIAAA Data Archive (Release 8Y, collection 4513, https://nda.nih.gov/edit_collection.html?id=4513). These data are publicly available via the provided links. The data used to generate

individual figures are provided in the Source Data files. Source data are provided with this paper.

Code availability

All original analysis code generated for this study is available on github through¹⁶³. In addition, code for computing the T2*-based estimates <https://doi.org/10.5281/zenodo.19631418> of basal ganglia tissue iron is available on github through <https://doi.org/10.5281/zenodo.8302458>. Original study code integrated publicly available software, including AFNI (available at <https://afni.nimh.nih.gov/pub/dist/doc/htmldoc/index.html>; version 23.1.10). All statistical analyses were conducted in R version 4.1 (<https://cran.r-project.org/bin/macosx/>).

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Competing interests

The authors declare no competing interests.

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Author contributions

A.C.P., F.J.C., S.F.T., K.N., D.B.G., D.C. and B.L. designed the study; D.F., D.B.G. and D.C. collected the data; A.C.P., A.O., D.J.P., F.J.C., B.T.C., W.F. and W.T. analyzed the data; A.C.P. drafted the manuscript with input from A.O., D.J.P., F.J.C., B.T.C., S.F.T., D.B.G., D.C. and B.L.

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